

Olga Petrovska, SFHEA

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Personal Statement

I am a dedicated educator with **9 years of experience** in supporting, coordinating, designing, and delivering a variety of undergraduate and postgraduate Computer Science modules. I have **expertise in computing education research** and **theoretical computer science** (logic, program extraction). My current research focuses on computing education practice, in particular assessments and the impact of GenAI in Higher Education. As a **Senior Fellow of the Higher Education Academy** with specialised **external examiner training**, I can provide thoughtful, impartial insights that enhance academic quality and student success.

Employment

- 2022 – 📌 **Head of Final Year** at Computer Science Department, Swansea University, UK
- Under my leadership (Interim Programme Director in 2022/23), our Applied Software Engineering programme achieved the best NSS scores in its history: 87% in learning opportunities, 88% in assessment and feedback, 98% in teaching, 100% in academic support
 - Developed a new marking system for final year projects that enables supervisors to quickly identify and resolve marking discrepancies, improving the speed and accuracy of feedback provided to students
 - Collaborate with external examiners and industry partners to balance academic standards within the programme with practical skills that enhance graduate employability
 - Lead the assessment review process for the whole programme, ensuring rigorous moderation and adherence to academic standards
- 2020 – 📌 **Teaching Fellow** at Computer Science Department, Swansea University, UK
- 2020 – 2025 📌 **Technology Consultant** and **Project Manager** at Ukraïner.net, remote (*part-time*)
- 2016 – 2020 📌 **Teaching Assistant** at Swansea University, UK (*part-time*)
- 2013 – 2015 📌 **Project Manager / Team Leader** at Wolfestone, Swansea, UK

Positions of Responsibility and Academic Environment

- 2025 – 📌 **Leader** of the “Embracing AI in Computer Science” Working Group within Swansea University’s CoSTA (Computer Science Teaching Attainment Group)
- 📌 **Co-leader** of the ITiCSE Working Group on Refactoring Upper Computing Courses to Leverage GenAI
- 📌 **Program Committee Member** at Human-Centered AI Education & Practice Conference (HCAI-ep’26), Computing Education Practice (CEP 2026), and Continuity, Computability, Constructivity (CCC 2025)
- 📌 **Reviewer** of HEA fellowship applications under Swansea University’s Advance HE-accredited Celebration of Professional Recognition (CoPR) programme
- 2024 – 📌 **Reviewer** for the several international journals (Frontiers in Education, Information and Software Technology, IEEE Transactions on Learning Technologies, Media and Communication, and IEEE Access) and conferences (SIGCSE TS 2026, CHI’26, OpenEd25, Koli Calling’25, and BCSWomen Lovelace Colloquium)
- 2023 – 📌 **Council Member** at the Association of Computability in Europe
- 📌 **Project Support Officer** at the UK-Ukraine Academic Mentorship Scheme
- 2020 – 📌 **Treasurer** at the British Colloquium for Theoretical Computer Science

Education

- 2016 – 2021  **Ph.D., Computer Science**
Swansea University, UK
Thesis: *Enhanced Realizability Interpretation for Program Extraction*
- 2015 – 2016  **M.Sc., Computer Science**
Swansea University, UK
Thesis: *Prawf: Interactive Proof Assistant for Predicate Logic*
- 2006 – 2007  **Master of Philology, English Language and Literature (Teaching Track)**
Chernivtsi National University, Ukraine
Thesis: *The Functional-Semantic Field of Quantity in John Fowles' Discourse*
- 2002 – 2006  **Bachelor of Pedagogy, Pedagogical Education**
Ternopil National Pedagogical University, Ukraine

Teaching Experience and Course Design

- 2025 –  Design and delivery of **CSB120MC Software Testing**
 Design and delivery of **CSF1007 Professional Issues**
- 2021 –  Design and delivery of **CSF302 Project Planning and Management**
 Design and delivery of **CSF206 Algorithms and Automata**
 Coordination of final year project modules: **CSF301 Project Specification and Development** and **CSF300 Project Implementation and Dissertation**
 Module support for **CS-170/CS-175 Modelling Computing Systems**
- 2024 – 2025  Co-designed **CSF325 Artificial Intelligence**
- 2021 – 2025  Designed and delivered **CSF107 Professional Issues of Software Engineering**
 Designed and delivered a range of short courses and workshops for teachers, college students, and general public on **Artificial Intelligence, Cyber Security, Graphic Design, Web Development, Excel**, and other.
- 2022 – 2023  Delivered **CSB100MC Computational Thinking**
 Delivered **CSB300MC Software Engineering Project Management**
 Delivered components of **CS-175 Modelling Computing Systems 2**
 Delivered components of **CSC301/CSCM01 Software Engineering Project Planning and Management**
- 2021– 2022  Delivered components of **CSF399 Human-Computer Interaction**
- 2020 – 2021  Delivered components of **CSF207 Computer Security**
 Designed and delivered **CSF106 Discrete Mathematics for Computer Science**
- 2016 – 2020  Provided support with labs, tutorials and assessment in a variety of CS modules:
- **CS-205 Declarative Programming**
 - **CSCM13 Critical System / CSC313 High Integrity Systems**
 - **CSCM75 Logic in Computer Science**
 - **CS-275 Automata Theory and Formal Languages**
 - **CS-o81 Introduction to Algorithms and Data Structures**

Publications

Conference Proceedings

- 1 D. J. Bouvier, B. P. Cipriano, R. Glassey, R. Pettit, **O. Petrovska**, E. Anderson, A. Birillo, R. Dougherty, N. Pombo, E. Rahimi, C. Ramakrishnan, A. Steinmaurer, S. Taneja, M. Usman, and A. Vadaparty, "The Rest of the Robots: Generative AI in Post-introductory Computing Education," in *2025 Working Group Reports on Innovation and Technology in Computer Science Education*, ser. ITiCSE 2025, Association for Computing Machinery, forthcoming.
- 2 **O. Petrovska** and C. Hopkins, "Propaganda or Exposure? A Video-Based Group Assessment Exploring Ethics in a Fictional Technological Society," in *Proceedings of the 10th Conference on Computing Education Practice*, ser. CEP'26, Durham, UK: Association for Computing Machinery, Jan. 2026, ISBN: 979-8-4007-2121-2/26/01.  DOI: 10.1145/3772338.3772352, in press.
- 3 O. Andrei, S. W. Nabi, M. Barr, and **O. Petrovska**, "Integrating Socially Responsible Computing Competencies in Computer Science and Software Engineering Education," in *Proceedings of the 9th Conference on Computing Education Practice*, ser. CEP'25, Association for Computing Machinery, Jan. 2025, pp. 36–37, ISBN: 9798400711725.  DOI: 10.1145/3702212.3702225.
- 4 D. J. Bouvier, B. P. Cipriano, R. Glassey, R. Pettit, E. Anderson, A. Birillo, R. Dougherty, O. Hazzan, **O. Petrovska**, N. Pombo, E. Rahimi, C. Ramakrishnan, A. Steinmaurer, S. Taneja, M. Usman, A. Vadaparty, and G. R. Yeluripati, "GenAI Integration in Upper-Level Computing Courses," in *Proceedings of the 30th ACM Conference on Innovation and Technology in Computer Science Education V. 2*, ser. ITiCSE 2025, Nijmegen, Netherlands: Association for Computing Machinery, Jun. 2025, pp. 691–692, ISBN: 9798400715693.  DOI: 10.1145/3724389.3731276.
- 5 L. Clift and **O. Petrovska**, "Learning without Limits: Analysing the Usage of Generative AI in a Summative Assessment," in *Proceedings of the 9th Conference on Computing Education Practice*, ser. CEP'25, Association for Computing Machinery, Jan. 2025, pp. 5–8, ISBN: 9798400711725.  DOI: 10.1145/3702212.3702214.
- 6 **O. Petrovska**, L. Clift, and F. Pantekis, "Video in Assessments for Soft Skill Development and Evaluation," in *Proceedings of the 2025 Conference on UK and Ireland Computing Education Research*, ser. UKICER'25, Association for Computing Machinery, Sep. 2025, ISBN: 9798400720789.  DOI: 10.1145/3754508.3754531.
- 7 **O. Petrovska**, R. Pearsall, and L. Clift, "Assessing Software Engineering Students' Analytical Skills in the Era of Generative ai," in *Proceedings of the 9th Conference on Computing Education Practice*, ser. CEP'25, Association for Computing Machinery, Jan. 2025, p. 34, ISBN: 9798400711725.  DOI: 10.1145/3702212.3702223.
- 8 **O. Petrovska**, L. Clift, F. Moller, and R. Pearsall, "Incorporating Generative AI into Software Development Education," in *Proceedings of the 8th Conference on Computing Education Practice*, ser. CEP'24, Durham, United Kingdom: Association for Computing Machinery, Jan. 2024, pp. 37–40, ISBN: 9798400709326.  DOI: 10.1145/3633053.3633057.
- 9 **O. Petrovska**, L. Clift, and F. Moller, "Generative AI in Software Development Education: Insights from a Degree Apprenticeship programme," in *Proceedings of the 2023 Conference on United Kingdom & Ireland Computing Education Research*, ser. UKICER'23, Swansea, Wales UK: Association for Computing Machinery, Sep. 2023, ISBN: 9798400708763.  DOI: 10.1145/3610969.3611132.
- 10 U. Berger, **O. Petrovska**, and H. Tsuiki, "Prawf: An Interactive Proof System for Program Extraction," in *Beyond the Horizon of Computability*, M. Anselmo, G. Della Vedova, F. Manea, and A. Pauly, Eds., Cham: Springer International Publishing, 2020, pp. 137–148, ISBN: 978-3-030-51466-2.  DOI: 10.1007/978-3-030-51466-2_12.
- 11 U. Berger and **O. Petrovska**, "Optimized Program Extraction for Induction and Coinduction," in *Sailing Routes in the World of Computation*, F. Manea, R. G. Miller, and D. Nowotka, Eds., Cham:

Springer International Publishing, 2018, pp. 70–80, ISBN: 978-3-319-94418-0.  DOI:
10.1007/978-3-319-94418-0_7.

Other

- 1 **O. Petrovska**, *Course Portfolio on Algorithms and Automata (Apprenticeship)*, 2024.  DOI:
10.5281/zenodo.14448266.
- 2 **O. Petrovska**, *Enhanced Realizability Interpretation for Program Extraction*, 2021.  DOI:
10.23889/suthesis.57831.

Event Organisation

- Jun 2026  **Logic Colloquium 2026** (upcoming) – Local organiser¹
- Sep 2025  **CCC 2025** (Continuity, Computability, Constructivity) – Local organiser
- Mar 2025  **UK-Ukraine Research Twinning Showcase Event 2025** at the Liverpool University – Co-organiser
- Jul 2024  **CCA 2024** (International Conference on Computability and Complexity in Analysis) – Local organiser
- May 2024  **Agda Implementors' Meeting XXXVIII** – Local organiser
- Sep 2023  **UKICER 2023** (UK and Ireland Computing Education Research) – Local organiser
- Jul 2022  **CiE 2022** (Computability in Europe: Revolutions and Revelations in Computability) – Local organiser
- Apr 2022  **BCTCS 38** (British Colloquium for Theoretical Computer Science) – Co-chair
- Apr 2020  **BCTCS 36** online conference – Local organiser
- Sep 2019  **2nd International Summer School on Proof Theory and Workshop on Proof Theory and its Applications** – Local organiser

Secondments and Residences

- Jul-Aug. 2018  **Residence** at the Hausdorff Trimester Program (Hausdorff Research Institute for Mathematics, Germany)
- Mar-Apr. 2018  **Secondment** at the Mathematical and Information Sciences Division of the Human and Environmental Studies Faculty (Kyoto University, Japan)
- Mar-May. 2017  **Secondment** at the School of Mathematics and Statistics (University of Canterbury, New Zealand)

CPD Courses and Certifications

- 2025  **Personal Development Course for External Examiners**, Advance HE
-  **Level 4 NVQ Diploma in Management**, ILM
-  **Project Management Professional Certificate**, Google
-  **Sustainability in the Digital Age: Energy-Efficient Software Development**, openHPI
-  **Sustainability in the Digital Age: Environmental Impacts of AI Systems**, openHPI
- 2024  **AI in Education: Leveraging ChatGPT for Teaching**, University of Pennsylvania
- 2023  **Data Science Bootcamp**, openHPI
- 2022  **Introduction to Quantum Computing with Qiskit (with IBM Quantum)**, openHPI

¹All conferences/events organised at Swansea University, unless stated otherwise.

CPD Courses and Certifications (continued)

- **Sustainable Software Engineering**, openHPI
- **Cyberthreats by Malware**, openHPI
- **Confidential Communication in the Internet**, openHPI

Selected Talks

- 1 **The Use of Video in Assessments for Soft Skill Development and Evaluation** at the University of Nottingham (26.11.2025, online)
- 2 **Smile! You're Being (Video) Assessed** at the Teaching Hour of the Computer Science Education Group, University of Edinburgh (24.11.2025, online)
- 3 **Modular Generative AI Training to Promote AI Literacy Across the University** at the Swansea University annual Learning and Teaching Conference SUSALT 2025 - Enhancing Our Practice: Share - Learn - Create | co-presented with Filippas Pantekis (10.07.2025, Swansea University, Wales)
- 4 **The Art of Teaching Theory Across Diverse Backgrounds** at the British Colloquium for Theoretical Computer Science 2025 (14-16.04.2025, University of Strathclyde, Scotland)
- 5 **GenAI in Education and Research** at the NAIADES (Network for Artificial Intelligence, the Arts, the Digital Economy and Society) Workshop "GenAI: Use and Best Practices" (20.02.2025, Swansea University, Wales)
- 6 **Computer Science Outreach in Higher Education: Challenges and Opportunities** | co-presented with Megan Venn-Wycherley (08.01.2025, Durham University, England)
- 7 **Developing Students' Critical Skills in the AI-enabled World** at the Advance HE Assessment and Feedback Symposium 2024 (05.11.2024, online)
- 8 **How are students using Generative Artificial Intelligence?** (01.11.2024, "Pinch of SALT" podcast)
- 9 **Generative AI and Conversations with Your Students** at the Lunch and Learn Webinar of the Swansea Academy of Learning and Teaching | co-presented with Lee Clift (22.10.2024, online)
- 10 **Student vs AI** at the Learning and Teaching Enhancement Centre (LTEC) Conference "GenAI: Challenges and Solutions" (18.09.2024, Swansea University, Wales)
- 11 **Operational Semantics in Prawf** at the Workshop on Proof and Computation 2022 (01.06.2022, Kloster Schlehndorf, Germany)
- 12 **Intuitionistic Fixed Point Logic and Program Extraction** at the Workshop on Continuity, Computability, Constructivity 2018 (24-28.09.2018, University of Algarve, Faro, Portugal)
- 13 **Natural Language Proofs for Program Extraction** at the International Workshop on Proof, Computation, Complexity 2017 (26-27.07.2017, University of Göttingen, Germany)
- 14 **Natural Language Proof Checking in Predicate Logic** at the Workshop on Proofs, Programs, and Verification (19.04.2017, University of Canterbury, Christchurch, New Zealand)

References

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